7C
GASES

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 7

1. Define gas pressure and its units.
2. Calculate the number of moles, pressure, volume, or temperature in a sample of gas using the quantities of the other three.
3. Calculate the properties of each gas in a mixture of gases.
4. Predict how the pressure in a fixed-volume container will change after a complete reaction.
5. Use the kinetic molecular theory to explain the behavior of gases.
6. Describe the relationship between molecular mass and speed.
7. Calculate the relative speeds and molecular masses of gases using root mean square speed and Graham's law of effusion.

Kinetic Molecular Theory of Gases

7.10

<u>Kinetic-molecular theory (KMT)</u>: A model (theory) that explains the observed macroscopic properties of gases, the ideal gas law equation, and the behavior of gas molecules at a microscopic scale.

<u>Five postulates of KMT:</u>

1. Gases are composed of small particles that are in constant, random motion.
2. The volume of the gas particles themselves is insignificant compared with the overall volume of the container.
3. Forces between the gas particles are negligible, except when the particles collide with one another.
4. Particle collisions are *perfectly* elastic; that is, no energy is lost when the particles collide.
5. The *average kinetic energy* (KE_{avg}) of the gas particles is directly proportional to the absolute temperature (Kelvin, K) of the gas sample.

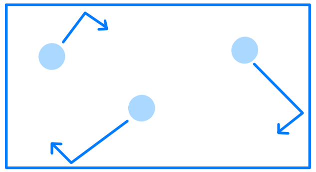
Kinetic Molecular Theory of Gases

7.10

Using KMT to explain gas pressure:

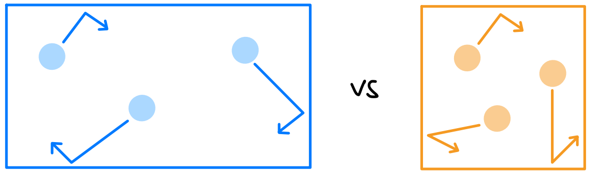
- Postulates 1 and 3 state that gas particles are in constant motion and are colliding with each other.
- The particles must also be colliding with the walls of the container.
- The force of collisions with walls is pressure.

$$P = \frac{F}{A}$$

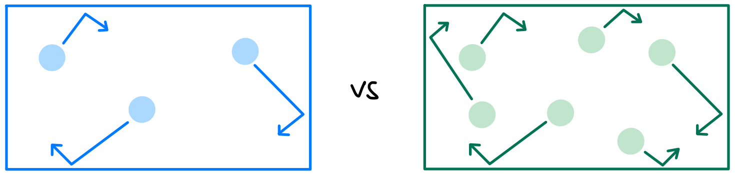


Q How to increase pressure?

$V\downarrow$ why? more frequent collisions with walls



$n\uparrow$ why? more particles, so more collisions with walls



$T\uparrow$ why? particles move faster, so more frequent collisions with walls

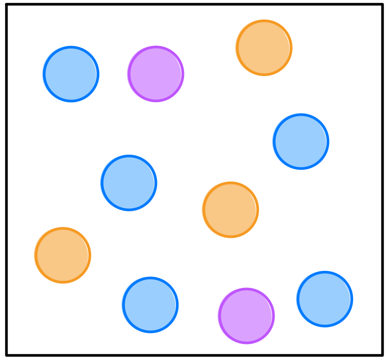
Kinetic Molecular Theory of Gases

7.10

<u>Using KMT to explain Dalton's law of partial pressures</u>:

- All the postulates of kinetic molecular theory consider only the numbers (or moles) of gas particles and not the identities of the gases.
- The partial pressure of a gas in a mixture is related only to the number of moles of each gas.

P_B, P_O, P_P

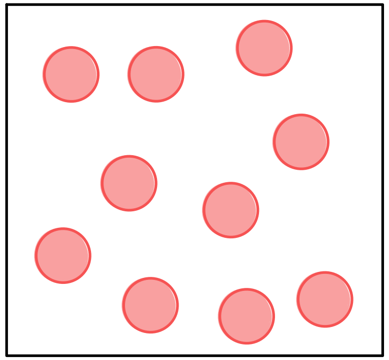


all same V, T

$$P_{tot} = \frac{(n_B + n_O + n_P)RT}{V}$$

Same $n_{tot} = 10$,
Same V and T,
so P_{tot} same!

$$P_{tot} = \frac{(n_{tot})RT}{V}$$



all same V, T

Movement of Gas Particles

7.11

Postulate 5: The average kinetic energy (KE_{avg}) of the gas particles is directly proportional to the absolute temperature (Kelvin, K) of the gas sample.

- Any two gas samples at the same temperature have the same KE_{avg}.

$T \propto KE_{avg}$; $KE_{avg} = \frac{1}{2} m \cdot v_{rms}^2$

The **root mean square speed** (v_{rms}) is the velocity of a gas particle possessing the average kinetic energy (KE_{avg}) in a sample of gas.

units: $\frac{m}{s}$

$$v_{rms} = \sqrt{\frac{3RT}{MM}}$$

$R = 8.314 \frac{J}{mol \cdot K}$

still gas constant, but different units

T in Kelvin (K)

MM = molar mass in $\frac{kg}{mol}$

Movement of Gas Particles

7.11

The <u>root mean square speed</u> (v_{rms}) is dependent on both the temperature and mass.

$v_{rms} \propto \frac{1}{mass}$ as mass ↓, speed ↑

$v_{rms} \propto T$ as temperature ↑, speed ↑

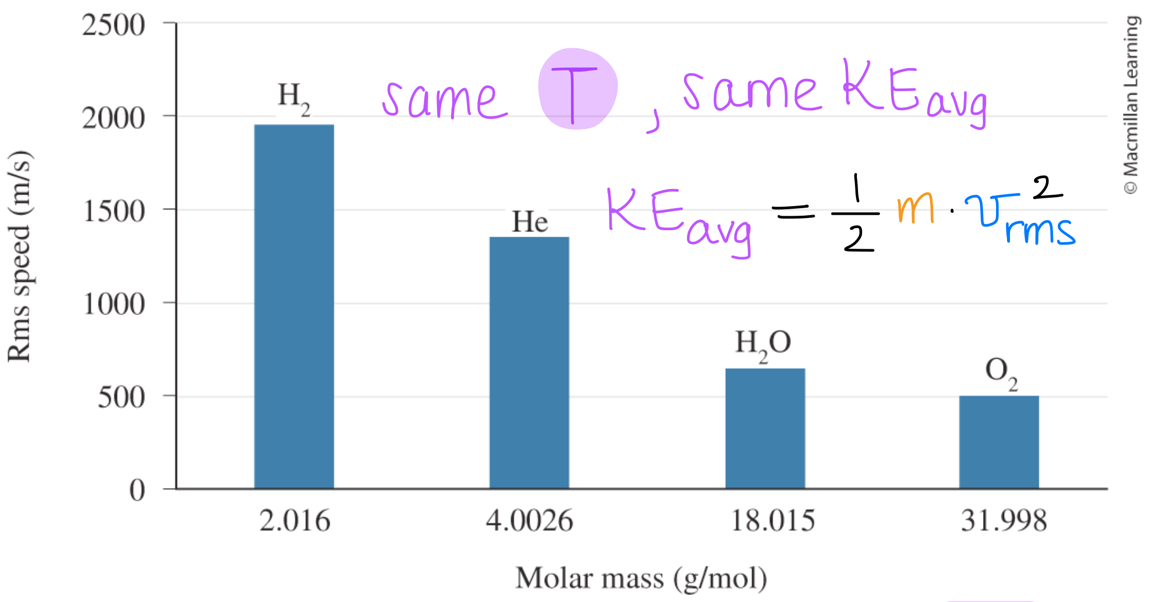


FIGURE 7.23 RMS Speeds of different gases at 25 °C

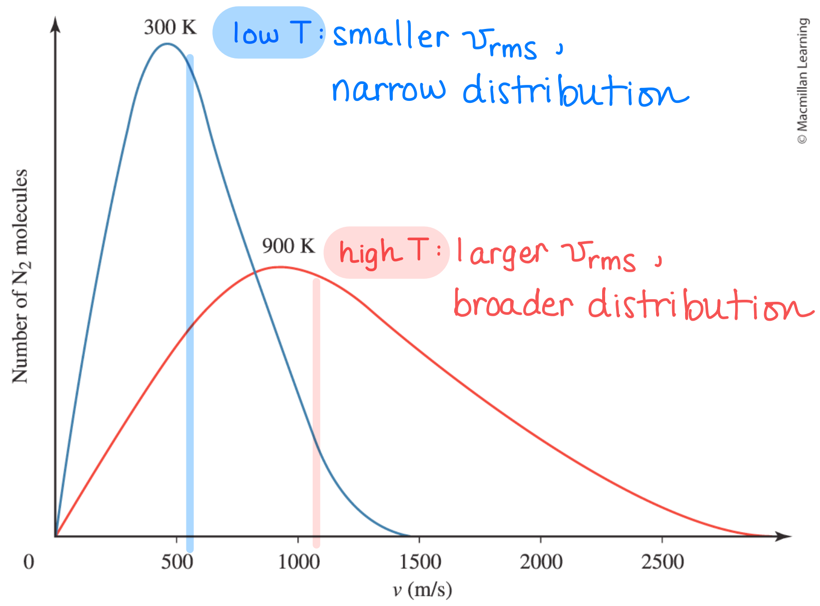


FIGURE 7.24 Distribution of speeds for N_2 gas at high and low T

Movement of Gas Particles

7.11

Diffusion: The process by which gas molecules spread out in response to a concentration gradient (i.e. mixing of gases).

- Influenced by root mean square speed (mass and temperature)
- Heavier molecules diffuse more slowly than lighter ones

$$v_{rms} = \sqrt{\frac{3RT}{MM}} \quad \text{or} \quad v_{rms} \propto \sqrt{\frac{T}{MM}}$$

rate $\uparrow$ as T $\uparrow$ because $v_{rms} \uparrow$

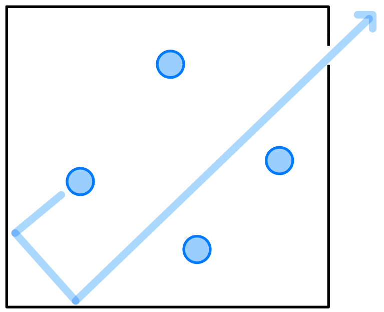
rate $\uparrow$ as MM $\downarrow$ because $v_{rms} \uparrow$

Movement of Gas Particles

7.11

Effusion: The process by which a gas escapes from a container (often, into a vacuum) through a small hole.

- Influenced by root mean square speed (mass and temperature)
- Heavier molecules effuse more slowly than lighter ones



$v_{rms} \propto \sqrt{\frac{T}{MM}}$

rate $\uparrow$ as $T \uparrow$ because $v_{rms} \uparrow$

rate $\uparrow$ as $MM \downarrow$ because $v_{rms} \uparrow$

Graham's Law of Effusion two gases (A, B) at same T :

$$\frac{rate_A}{rate_B} = \sqrt{\frac{MM_B}{MM_A}}$$

EXAMPLE PROBLEM

Learning Objective(s): 7.7

A sample of hydrogen gas, H_2, was found to effuse through a pinhole 5.2 times as rapidly as an unknown gas (at the same temperature, pressure, and volume).

X X

What is the molecular weight of the unknown gas?

this means H_2 is faster,
so X must be slower and heavier

Graham's Law: $\dfrac{r_{H_2}}{r_x} = \dfrac{\sqrt{MM_x}}{\sqrt{MM_{H_2}}}$

$$\frac{5.2}{1} = \frac{\sqrt{MM_x}}{\sqrt{2.016\ g/mol}}$$

$$\sqrt{MM_x} = \frac{5.2}{1} \cdot \sqrt{2.016\ g/mol}$$

$$\sqrt{MM_x} = 7.3_8 \sqrt{g/mol}$$

$$\boxed{MM_x = 55\ g/mol}$$

H_2 faster, so X must be heavier!

IN-CLASS QUESTION 1

Learning Objective(s): 7.5

Which of the following statement(s) is/are true of ideal gases?

- ☑ No energy is lost when molecules of ideal gases collide.
- ☑ Molecules of F_2 gas and Cl_2 gas have the same average kinetic energy at 875 K.

  same T, same KE_{avg}
  $T \propto KE_{avg}$
- ☒ Particles of gas are more attracted to one another at 1000 K than at 300 K because the particles at 1000 K have greater kinetic energy.
- ☑ Kinetic molecular theory adequately explains why increasing the number of moles of gas in a sample increases pressure. due to more frequent collisions with walls of container
- ☒ According to kinetic molecular theory, pressure increases as volume decreases because the particles are ~~*more attracted* to the walls of the container~~, leading to more forceful collisions.

IN-CLASS QUESTION 2

Learning Objective(s): 7.5, 7.6

different molar mass !

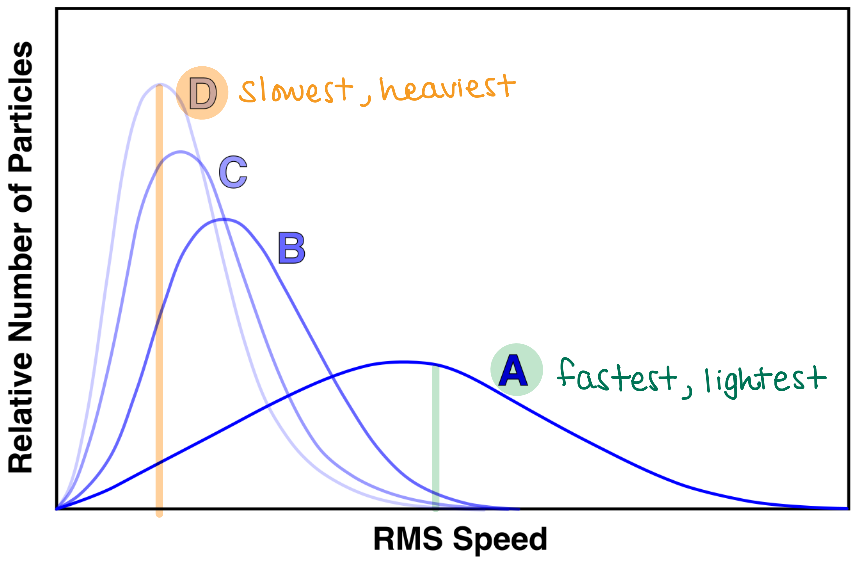
Same KE_{avg}

The following graph represents four different gases (**A–D**) at the *same temperature*.

Which of the following statements are true?

$v \propto \frac{1}{\text{mass}}$

- ☒ Molecules of gas A have a greater distribution of speeds due to the gas having a larger molar mass.
- ☒ Molecules of gas C have a smaller molar mass than the molecules of gas B.
- ☒ All of the curves actually represent the same compound at the same temperature, but have different root mean square speeds.
- ☒ Molecules of gas D have the *smallest* root mean square speed and would diffuse the fastest.
- ☑ Molecules of gas B will effuse faster than molecules of gas C.



IN-CLASS QUESTION 3

Learning Objective(s): 7.7

It takes a sample of of Ne(g) 7.40 minutes to effuse a certain distance.

How long (in minutes) will it take for the same amount of Br_2(g) to effuse the same distance under the same temperature and pressure?

Graham's Law: $\dfrac{r_{Ne}}{r_{Br_2}} = \dfrac{\sqrt{MM_{Br_2}}}{\sqrt{MM_{Ne}}} = \dfrac{\sqrt{159.808\ g/mol}}{\sqrt{20.180\ g/mol}}$

$$\frac{r_{Ne}}{r_{Br_2}} = 2.81409$$

Br_2 heavier, so it's slower

$$\text{rate} = \frac{\text{distance}}{\text{time}} \qquad r_{Ne} = \frac{x}{t_{Ne}} \qquad r_{Br_2} = \frac{x}{t_{Br_2}}$$

$$\frac{r_{Ne}}{r_{Br_2}} = \frac{\frac{x}{t_{Ne}}}{\frac{x}{t_{Br_2}}}$$

$$\frac{r_{Ne}}{r_{Br_2}} = \frac{t_{Br_2}}{t_{Ne}}$$

$$2.81409 = \frac{t_{Br_2}}{7.40\ \text{min}}$$

$$\boxed{t_{Br_2} = 20.8\ \text{min}}$$

BONUS QUESTION

Assessed on: Summer 2021, Exam 2

Which of the molecules (all in the gas phase) will have the *slowest* rate of effusion?

- ❑ F_2 at 315 K
- ❑ Br_2 at 298 K
- ❑ CO_2 at 310 K
- ❑ CH_4 at 325 K